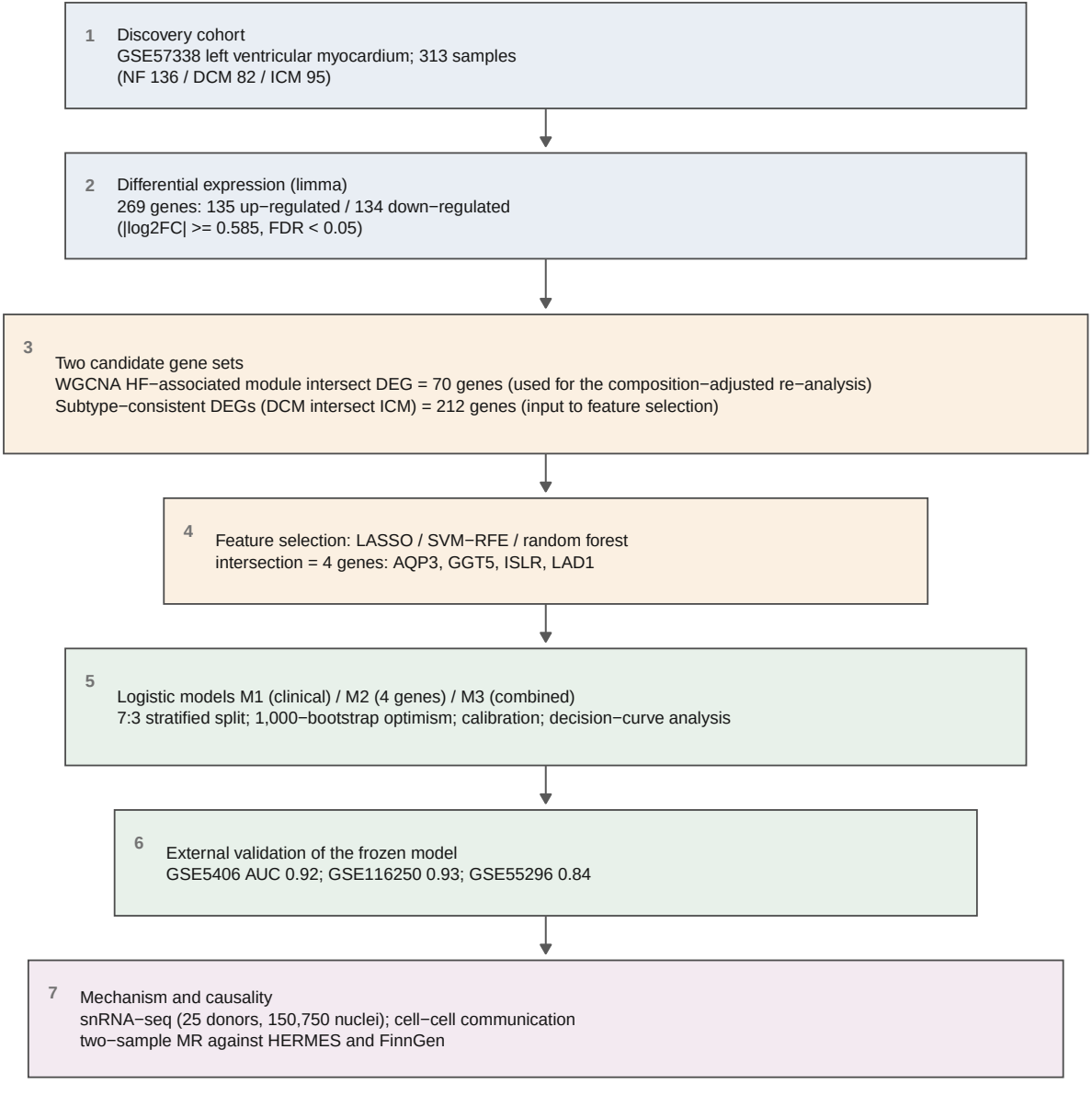
1
Discovery cohort
GSE57338 left ventricular myocardium; 313 samples
(NF 136 / DCM 82 / ICM 95)
2
Differential expression (limma)
269 genes: 135 up−regulated / 134 down−regulated
(|log2FC| >= 0.585, FDR < 0.05)
3
Two candidate gene sets
WGCNA HF−associated module intersect DEG = 70 genes (used for the composition−adjusted re−analysis)
Subtype−consistent DEGs (DCM intersect ICM) = 212 genes (input to feature selection)
4
Feature selection: LASSO / SVM−RFE / random forest
intersection = 4 genes: AQP3, GGT5, ISLR, LAD1
5
Logistic models M1 (clinical) / M2 (4 genes) / M3 (combined)
7:3 stratified split; 1,000−bootstrap optimism; calibration; decision−curve analysis
6
External validation of the frozen model
GSE5406 AUC 0.92; GSE116250 0.93; GSE55296 0.84
7
Mechanism and causality
snRNA−seq (25 donors, 150,750 nuclei); cell−cell communication
two−sample MR against HERMES and FinnGen